From Drawing Product to Drawing Process: Combining 2D Sketch Analysis with Video-Based 3D Hand Kinematics for Parkinson's Motor Assessment

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Abstract—Parkinson's disease (PD) affects over 10 million people worldwide, with motor symptoms commonly assessed through drawing and handwriting tasks. Current digital assessment requires specialized tablets costing $500–$2000, limiting at-home monitoring. We present a two-part feasibility study combining 2D drawing product analysis with 3D video-based hand kinematic process analysis for PD motor assessment.

In Part A, we analyze a public dataset of 204 spiral and wave drawings contributed by 28 people (15 with PD, 13 healthy), treating the person as the unit of analysis. Image entropy was higher in PD (Cohen's $d = 0.80$, $p = 0.042$; not significant after Holm correction across 16 features), and person-grouped cross-validation reached 60–62% accuracy (chance ≈50%). The finished drawing alone therefore carries only a modest signal, which motivates looking at how the drawing is made.

In Part B, we use smartphone video-based 3D hand tracking (MediaPipe) to analyze hand kinematics during the drawing act. In a proof-of-concept with 2 healthy participants performing spiral tracing and sentence writing under simulated PD conditions (mild tremor, strong tremor, restricted movement, micrographia), we observe that: (1) under simulated mild tremor, tremor amplitude, peak speed, normalized jerk, speed variability and 4–6 Hz band power increased in both participants; (2) during simulated micrographia, movement smoothness (SPARC) fell in both participants (-1.67 and -0.85), writing took longer and mean speed was lower, without analyzing the written text. With $n = 2$, these are descriptive, hypothesis-generating observations, not statistical estimates.

Together, the two parts outline a framework in which process analysis, requiring only a standard smartphone camera, complements product analysis; clinical validation with diagnosed patients is the necessary next step.

Index Terms—Parkinson's disease, hand tracking, sketch analysis, motor assessment, tremor detection, MediaPipe, smartphone sensing, micrographia, SPARC, rehabilitation monitoring

I. Introduction

Parkinson's disease (PD) is the second most common neurodegenerative disorder, affecting over 10 million people worldwide [1]. The cardinal motor symptoms—resting tremor (4–6 Hz), bradykinesia, rigidity, and postural instability—progressively impair fine motor control. Handwriting and drawing tasks are among the earliest and most sensitive clinical indicators of motor decline [2].

Clinical assessment relies on the Unified Parkinson's Disease Rating Scale (UPDRS) [3], administered during clinic visits every 2–4 months. Between visits, patients lack objective motor monitoring, making medication adjustment reactive. Drawing tasks—particularly the Archimedes spiral test and handwriting analysis for micrographia—are standard clinical tools [4], yet their scoring remains largely subjective or requires specialized tablets costing $500–$2000 [5].

Two fundamentally different approaches exist for computational drawing analysis: (1) product analysis—examining the final drawn output, and (2) process analysis—examining how the drawing was produced. Product analysis has been extensively studied [5], [6]. Process analysis, which captures richer temporal and kinematic information, has been limited to specialized hardware. Recent advances in markerless hand tracking— particularly Google's MediaPipe framework [7], which provides real-time 3D estimation of 21 hand joint positions from monocular RGB video—now enable process analysis using only a standard smartphone camera.

Research Gap. While 2D drawing product analysis for PD is well-established, to the best of our knowledge, no prior study has investigated: (1) whether video-based 3D hand kinematics during drawing can detect PD motor symptoms, or (2) how product and process analysis complement each other for comprehensive motor assessment.

Contributions. This paper presents:

1) A two-part framework combining 2D drawing product analysis with 3D video-based hand kinematic process analysis.
2) A person-level evaluation of 16 image-based sketch features on a public PD drawing dataset (204 drawings, 28 people), using person-grouped cross-validation so that no individual appears in both training and test data.
3) A 16-feature hand kinematic set computed from smartphone video, and pilot observations ($n = 2$) that these features respond in the expected direction

to simulated tremor, and that simulated micrographia is reflected in hand movement itself, without analyzing the written text.

4) A discussion of how process analysis could capture motor information that product analysis cannot, enabling future accessible, hardware-free PD monitoring, contingent on clinical validation.

II. Related Work

A. Drawing and Handwriting Analysis for PD

Drawing tasks have been used for PD motor assessment for decades [4]. The Archimedes spiral test is widely used to quantify tremor severity, with features such as drawing speed, smoothness, and deviation from template correlating with UPDRS motor scores [15]. Micrographia—abnormally small or progressively shrinking handwriting—is a common feature of PD and can precede clinical diagnosis [16].

Digital tablet-based systems have enabled quantitative analysis of pen dynamics including pressure, velocity, and jerk [5], [6]. Drotár et al. [5] demonstrated that kinematic and pressure features from digitized handwriting could distinguish PD patients from controls with over 80% accuracy, but these systems require specialized hardware unavailable in home settings.

B. Sketch Recognition and Computational Drawing Analysis

Computational analysis of the drawing process has a rich history in the sketch recognition community. Early work by Rubine [8] established gesture feature extraction from pen trajectories, and Gross [9] examined the ambiguity of hand-drawn marks as a fundamental property of freehand input. Hammond and Davis [11] introduced the LADDER language for formally describing and recognizing hand-drawn sketch primitives, enabling systematic analysis of drawing structure. Paulson and Hammond [12] advanced geometric sketch recognition with accurate primitive recognition and beautification, while Fonseca et al. [10] demonstrated robust sketch understanding from noisy hand-drawn input.

A particularly relevant line of work examines not just the sketch product but the drawing process as a signal of cognitive and motor ability. Hammond et al. [13] proposed metrics for modeling expert sketching ability that capture fluency, temporal dynamics, and process characteristics—features conceptually parallel to our motor health assessment approach. Separately, rehabilitation systems such as Kinohaptics [14] have shown how continuous motor monitoring with automated haptic feedback can support post-surgical recovery, motivating similar approaches for neurodegenerative conditions.

C. Video-Based Hand Analysis

MediaPipe Hands [7] provides real-time 3D estimation of 21 hand landmarks from monocular video. While extensively applied to gesture recognition and sign language [17], its use for clinical motor assessment remains largely unexplored. Yan et al. [18] recently applied MediaPipe to hand tremor quantification. However, to the best of our knowledge, no prior study has combined video-based hand tracking with sketch-process analysis for PD motor symptom detection during functional drawing tasks.

III. Methodology

Our framework consists of two complementary approaches (Fig. 1): Part A analyzes the drawing product (what was drawn) using image processing on a public clinical dataset, while Part B analyzes the drawing process (how it was drawn) using video-based 3D hand tracking.

A. Part A: 2D Drawing Product Analysis

1) Dataset: We use the publicly available Parkinson's Drawings dataset [22], containing 204 scanned spiral and wave drawings (102 labelled PD, 102 healthy). The file names encode a participant code, from which we identified 28 contributors (15 with PD, 13 healthy), each contributing 6–30 drawings. Because drawings from the same person are not independent, the person is the unit of analysis throughout Part A.

2) Feature Extraction: We propose 16 image-based features:

- Tremor features: Contour smoothness (curvature variability), tremor intensity (high-frequency spectral power), stroke irregularity (deviation from smooth fit).
- Kinematic proxy features: Stroke density, speed proxy (inter-point distance), speed variability.
- Shape features: Drawing area, compactness, radial deviation, crossing count, Hu moments (1st and 2nd).
- Texture features: Stroke thickness variation, mean thickness, perimeter-area ratio, image entropy.

3) Statistical Analysis: For each person, each feature was averaged over all of their drawings. Healthy and PD groups were compared with Welch's t-test (Mann–Whitney U as a robustness check), with Holm correction across the 16 features.

4) Classification: Three classifiers were evaluated: Random Forest (100 trees, max depth 10), Gradient Boosting (100 estimators, max depth 5), and SVM with RBF kernel ($C = 1.0$, $\gamma = \text{scale}$). We used stratified group 5-fold cross-validation (StratifiedGroupKFold, grouped by person, 20 random repetitions), so that no person's drawings appear in both training and test folds. Z-score normalization was fitted within each training fold. As a second check, we ran leave-one-person-out validation and classified each held-out person by majority vote over their drawings.

B. Part B: 3D Video-Based Hand Kinematic Analysis

1) Participants and Setup: Two healthy participants (both male, ages 30 and 60) provided informal consent for this proof-of-concept study. A consumer smartphone (rear camera, 1080p at 30–60 fps) was mounted 20–25 cm

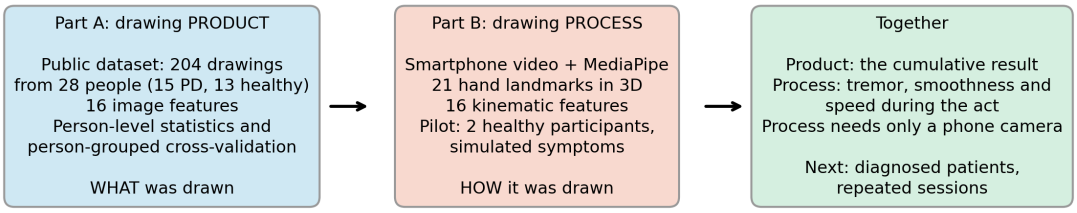


Fig. 1. Combined framework. Part A analyzes the 2D drawing product with image features, evaluated at the person level on a public dataset. Part B analyzes the 3D hand drawing process from ordinary smartphone video.

above the drawing surface. Participants drew on white A4 paper under consistent overhead lighting. No specialized hardware was used. With $n = 2$, this study is designed to establish technical and methodological feasibility, not to provide statistically generalizable estimates of effect size.

2) Drawing Tasks:

- Spiral tracing: Tracing a printed Archimedes spiral template (3 loops, 10 cm diameter).
- Sentence writing: Writing "Today is a beautiful day" on blank paper.

3) Simulated Symptom Conditions: To obtain kinematic reference ranges before clinical data collection, healthy participants performed the tasks under physically simulated conditions:

- Normal: No manipulation (baseline), for both tasks.
- Mild tremor (spiral): Vibrating motor device loosely secured near the drawing wrist to induce low-amplitude oscillation.
- Strong tremor (spiral): Same device firmly applied to increase oscillation amplitude.
- Restricted movement (spiral): Thick elastic wrap around the wrist to restrict range of motion. This is a kinematic perturbation related to reduced movement amplitude, not a physiological simulation of bradykinesia, which arises from distinct neuromotor mechanisms.
- Micrographia (writing): Participant instructed to write with progressively decreasing letter size [16].

Each recording lasted approximately 20–50 seconds. Total recordings: 12 videos (2 participants × 6 task–condition pairs). Every comparison below is made within the same task (spiral vs. spiral, writing vs. writing) and within the same person.

4) Hand Tracking Pipeline: Videos were processed using MediaPipe Hands [7], extracting 21 three-dimensional hand joint positions per frame. Detection and tracking confidence thresholds were set to 0.5. Mean detection rate across all videos was 64.0% (range: 10.7%–95.6%). Frames without a successful detection were not retained; only frames with valid landmarks were kept, each paired with its true elapsed timestamp. Velocity was computed as displacement divided by the actual (not assumed-uniform) time interval between consecutive retained frames, so detection gaps do not introduce artificial spikes into the derivative estimates. However, any rapid motion occurring within a detection gap is not captured and is instead averaged over the gap's true duration, which can understate peak velocity, jerk, or tremor content during low-detection-rate intervals (Section V-D).

5) Feature Extraction: We propose 16 hand kinematic features organized by PD symptom domain:

- Bradykinesia: Mean speed, speed variability, peak speed, normalized jerk [20], spectral arc length smoothness (SPARC) [19].
- Tremor: Tremor amplitude (3–12 Hz bandpass RMS), tremor frequency, PD-band power ratio (4–6 Hz).
- Stability: Grip stability (thumb-index distance variance), hand steadiness (wrist variability), wrist range of motion, finger coordination (inter-finger velocity correlation).
- Temporal: Drawing duration, pause ratio, velocity peaks per second, amplitude decay slope (micrographia index).

Let $\mathbf{p}_t \in \mathbb{R}^3$ denote the index fingertip position at frame t. The SPARC smoothness metric is computed as:

$$\text{SPARC} = -\int_0^{\omega_c} \sqrt{\left(\frac{1}{\omega_c}\right)^2 + \left(\frac{d\hat{V}(\omega)}{d\omega}\right)^2}\, d\omega \qquad (1)$$

where $\hat{V}(\omega)$ is the normalized Fourier magnitude spectrum of the velocity profile, and ω_c is the cutoff frequency (10 Hz) [19]. More negative SPARC values indicate less smooth movement.

6) Analysis: With two participants, Part B results are reported descriptively: for each person, the change from their own normal recording to the simulated condition in the same task. We report whether the direction of change agrees across both participants, and do not compute p-values.

TABLE I
Part A: Top Features at the Person Level (13 Healthy vs. 15 PD)

Feature	Healthy	PD	p	d
Image Entropy	5.18 ± 0.14	5.32 ± 0.21	0.042	0.80
Hu Moment 1	1.94 ± 0.09	2.02 ± 0.12	0.069	0.71
Crossing Count	45.8 ± 26.8	95.0 ± 95.5	0.074	0.70
Stroke Density	0.061 ± 0.013	0.073 ± 0.021	0.085	0.67
Speed Proxy	1.171 ± 0.011	1.176 ± 0.009	0.168	0.54

Per-person means of each feature, then mean $\pm$ SD across people. p: Welch's t-test (uncorrected; Holm-corrected $p \geq 0.68$ for all). d: Cohen's d, positive = higher in PD.

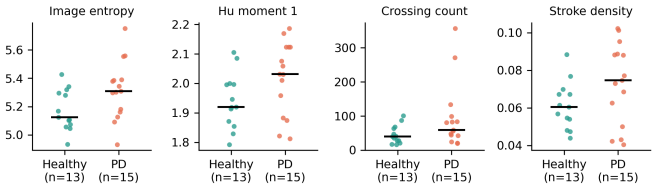


Fig. 2. Part A: one point per person (each person's drawings averaged). Bars show group medians. PD values tend to be higher, with substantial overlap.

IV. Results

A. Part A: 2D Drawing Product Analysis

1) Feature Differences: At the person level, image entropy was higher in PD ($p = 0.042$, Mann–Whitney $p = 0.038$, $d = 0.80$), but did not remain significant after Holm correction across 16 features. Hu moment 1, crossing count and stroke density were also higher in PD, with medium effect sizes ($d = 0.67$–0.71, $p = 0.07$–0.08) (Table I, Fig. 2).

2) Classification Results: Table II summarizes person-grouped classification accuracy. Across tasks and models, accuracy was 54–62%, against a chance level of about 50%. Leave-one-person-out results, classifying each held-out person by majority vote, ranged from 46% to 68%.

Overall, image features of the finished drawing carry a modest, consistent signal (PD drawings tend to be more irregular and denser), but on their own they do not separate people with PD from healthy people reliably in a 28-person sample.

B. Part B: 3D Video-Based Hand Kinematic Analysis

Note on interpretation: All results in this subsection derive from $n = 2$ participants and are reported as pilot, hypothesis-generating observations.

1) Hand Tracking Performance: MediaPipe tracked the hand in 64.0% of frames on average (range: 10.7%–95.6%). Higher detection rates coincided with better lighting and less motion blur. Ten of 12 videos had detection rates above 50%.

2) Simulated Tremor: Comparing each participant's normal spiral with their simulated mild-tremor spiral (Fig. 3), five of six tremor-related features increased

TABLE II
Part A: Classification Accuracy (%), Person-Grouped

Task	Model	Group 5-fold	LOPO (people)
Spiral	RF	59.9 ± 4.2	46.4
	GB	54.0 ± 5.0	57.1
	SVM	58.6 ± 3.9	46.4
Wave	RF	62.0 ± 2.9	64.3
	GB	59.1 ± 4.1	53.6
	SVM	61.9 ± 3.2	57.1
Combined	RF	62.2 ± 2.5	67.9
	GB	60.2 ± 3.1	64.3
	SVM	60.1 ± 2.4	60.7

Group 5-fold: StratifiedGroupKFold by person, mean $\pm$ SD over 20 repetitions (per drawing). LOPO: leave-one-person-out, majority vote per person ($n = 28$). RF: Random Forest; GB: Gradient Boosting; SVM: RBF kernel.

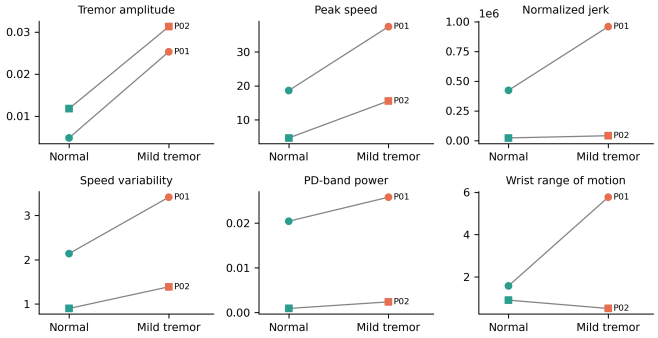


Fig. 3. Part B: normal vs. simulated mild tremor (spiral task), each line one participant. Five of six features rose in both participants.

in both participants: tremor amplitude (3–12 Hz RMS; +424% and +165%), peak speed (+101% and +233%), normalized jerk (+127% and +81%), speed variability (+60% and +54%), and PD-band power (4–6 Hz; +26% and +160%). Wrist range of motion increased in one participant (+267%) and decreased in the other (−44%). These directions are consistent with a tremor-like perturbation; with two participants, the magnitudes are not stable estimates.

3) Simulated Micrographia: Comparing each participant's normal sentence writing with their micrographia writing (Table III, Fig. 4), SPARC smoothness fell in both participants (−1.67 and −0.85), writing took longer in both (+19.2 s and +13.6 s), and mean speed was lower in both. The amplitude decay slope did not change consistently. These changes were measured from hand movement alone, without analyzing the written text, which would make this approach independent of language and of text visibility; confirming it requires diagnosed patients.

4) Restricted-Movement Observation: Under the restricted-movement condition, tremor-band amplitude was higher than the same person's normal spiral in both participants (0.079 vs. 0.005 and 0.087 vs. 0.012). SPARC did not change consistently, and one of these recordings

TABLE III
Part B: Micrographia vs. Normal Writing, per Participant

	P01		P02	
Feature	Normal	Micro.	Normal	Micro.
SPARC	−10.92	−12.59	−11.26	−12.11
Duration (s)	22.5	41.7	22.7	36.2
Mean speed	1.25	1.19	1.35	1.21

Writing task only. Lower SPARC = less smooth movement.

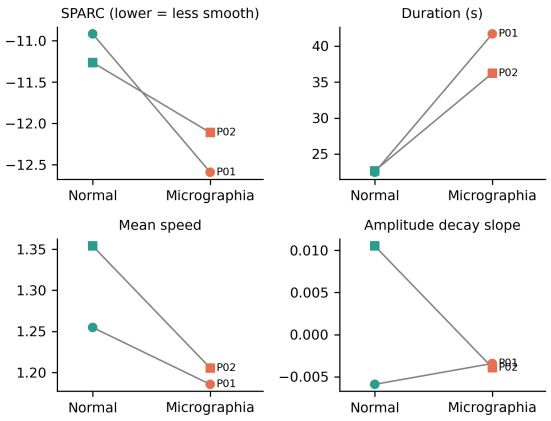


Fig. 4. Part B: normal vs. simulated micrographia (writing task), each line one participant. Smoothness, duration and speed changed in the same direction in both; amplitude decay did not.

had the lowest detection rate in the study (10.7%), so this condition is reported as an observation only; it approximates reduced movement amplitude rather than clinical bradykinesia.

V. Discussion

A. Complementary Nature of Product and Process Analysis

Product and process analysis capture different aspects of PD motor impairment. Product analysis (Part A) captures the cumulative effect on the final drawing. At the person level it gave a modest signal (60–62% accuracy), which is consistent with the fact that a finished drawing cannot show tremor that was motor-compensated, or hesitations and slowing that did not alter the final stroke. Process analysis (Part B), in principle, captures the moment-to-moment motor state during drawing—tremor, grip stability, smoothness, and velocity dynamics—potentially detecting motor phenomena before they manifest in the product. Our pilot shows kinematic features changing in the expected direction under simulated tremor and micrographia, an encouraging but preliminary signal of sensitivity. The combined framework's long-term promise is process analysis for daily monitoring at no additional hardware cost, alongside product analysis, contingent on Part B being validated at a much larger scale.

B. Key Findings

The finished drawing alone is a modest signal. Evaluated with the person as the unit, image features of the drawing showed consistent but small group differences and 54–62% classification accuracy. This sets a realistic baseline for product-only analysis on small public datasets and motivates adding information about how the drawing was produced.

Micrographia from hand kinematics (preliminary). In both participants, simulated micrographia was accompanied by less smooth, slower hand movement. If replicated in diagnosed patients, language-agnostic detection independent of text visibility would be clinically meaningful. The idea that how a task is executed may carry diagnostic signal beyond what is produced [13] motivates extending this principle to other motor assessment tasks beyond drawing.

Accessibility. Because Part B uses only a consumer smartphone rather than specialized tablet hardware, the approach has the potential to reduce hardware barriers to resource-limited or home-based motor monitoring, if the underlying approach is validated.

C. Broader Implications

Our framing of PD motor assessment as a sketch process analysis problem opens a productive connection to HCI and sketch recognition research. The kinematic features we extract—smoothness, velocity profiles, temporal dynamics, and spatial amplitude—are structurally similar to the features used to analyze drawing fluency and motor expertise in freehand input systems [8], [13]. Process analysis in Part B is conceptually equivalent to studying how someone draws rather than what they drew, a distinction central to sketch recognition systems [10], [11]. This analogy suggests that feature engineering advances from the sketch recognition literature could be applied to clinical motor assessment—a cross-disciplinary direction that warrants dedicated, larger-scale future investigation.

D. Limitations

- Part A sample size: Although the dataset contains 204 drawings, it represents 28 people, which limits statistical power; the medium effect sizes observed would need roughly 25–35 people per group to be confirmed. Participant identity was inferred from filename codes, and the drawings are scans rather than live recordings.
- Part B sample size: With $n = 2$ healthy participants, Part B is strictly proof-of-concept. We report directions of change, not statistical tests, and clinical validation with diagnosed patients is essential before any clinical claim can be made.
- Simulated vs. real PD: Physical symptom simulation in healthy participants does not replicate the neurological mechanisms of PD, and results may not transfer to real patients.

- Detection variability and gaps: Hand tracking ranged from 10.7% to 95.6% depending on lighting and motion blur. Frames without a successful detection were dropped rather than interpolated; velocity used the true elapsed time between retained frames, so gaps do not introduce artificial spikes, but rapid motion within a gap is averaged over its duration, which can understate peak velocity, jerk, or tremor content. Future work should add uncertainty-aware gap handling and validate tracking against a reference sensor (e.g., a wrist-worn accelerometer or a digitizing tablet).
- Individual baseline variability: Resting smoothness and tremor likely vary from person to person, so a fixed, group-derived threshold may under- or over-detect deviation for an individual. Establishing each person's own baseline before flagging deviation from it is likely necessary for reliable individual-level monitoring.

E. Future Work

Immediate priorities: (1) validating MediaPipe hand kinematics against a reference sensor and measuring test–retest reliability; (2) clinical validation with diagnosed PD patients (Hoehn & Yahr I–IV) and healthy controls at a sample size sufficient for confirmatory inference, including correlation with MDS-UPDRS motor scores; (3) larger public drawing datasets with more contributors for Part A; (4) longitudinal daily monitoring with personal baselines; (5) deep temporal models including state-space architectures [21] for hand trajectory classification; (6) unified product-plus-process feature fusion; (7) a real-time smartphone application providing clinician-interpretable output; and (8) extending the same kinematic pipeline toward interactive, immersive delivery—for example, virtual reality environments that gamify repeated drawing or writing exercises—which could support motor rehabilitation rather than measurement alone.

VI. Conclusion

We presented a two-part feasibility study of complementary approaches to Parkinson's motor assessment through drawing analysis. Part A, evaluated with the person as the unit of analysis (28 people, 204 drawings), found that image features of the finished drawing carry a modest signal: image entropy was higher in PD ($d = 0.80$) and person-grouped classification reached 60–62%. Part B, a small $n = 2$ pilot, found that 3D hand kinematics from an ordinary smartphone video changed in the expected direction under simulated symptoms: five tremor-related features rose in both participants under simulated tremor, and movement smoothness fell in both during simulated micrographia, without analyzing the written text.

The transition from drawing product to drawing process analysis represents a promising direction: from analyzing what patients drew to analyzing how they draw, using only a phone camera. Validation against reference sensors and with diagnosed PD patients will determine whether these proof-of-concept results generalize to real-world motor assessment and longitudinal monitoring.

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